DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2023-24
PRE BOARD EXAMINATION

CLASS: X | **FULL MARKS: 80**
SUBJECT: CHEMISTRY [SET A] | **TIME: 2 HOURS**

Answers to this paper must be written on the paper provided separately.
You will not be allowed to write during first 15minutes.
This time is to be spent in reading the question paper.
The time given at the head of this Paper is the time allowed for writing the answers.
Section A is compulsory. Attempt any four questions from Section B.
The intended marks for questions or parts of questions are given in brackets [ ].
(This question paper contains seven printed pages)

SECTION A

(Attempt all questions from this Section)

Question1

Choose the correct answers to the questions from the given options. **[15]**
(Do not copy the question, write the correct answers only)

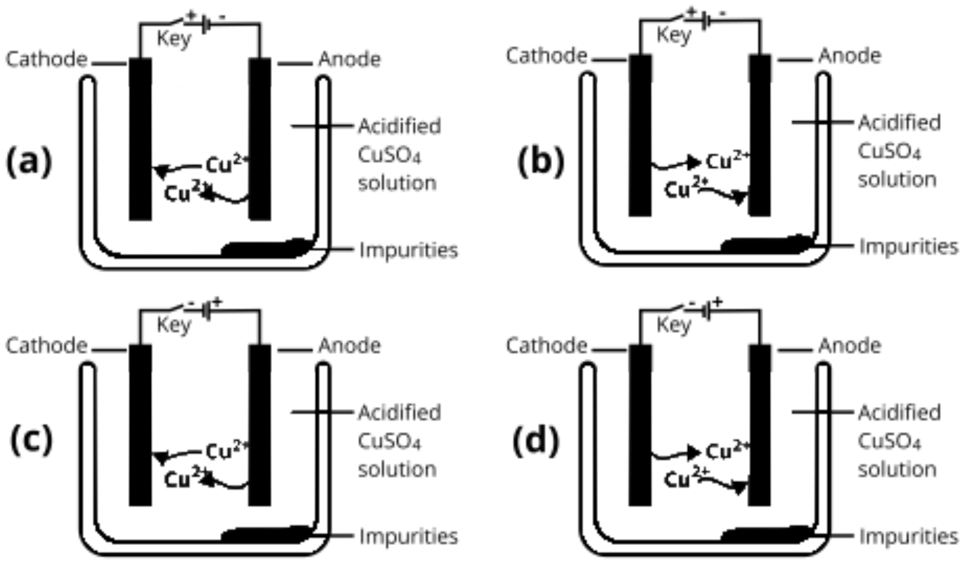
(i) **The atomic masses of sulphur, oxygen, and helium are approximately 32, 16 and 4 respectively. Which of the following statements regarding the number of molecules in 32 g of sulphur, 16 g of oxygen and 4 g of helium.**

M: 16 g of oxygen contains eight times the number of molecules as 4 g of helium.

N: 16 g of oxygen contains half the number of molecules as 32 g of sulphur.

(a) Both M and N
(b) Neither M or N
(c) N
(d) M

(ii) **Which of the following is correct description of the process of electrorefining of copper?**



(a) c
(b) d
(c) b
(d) a

(iii) A compound P is heated in a test tube with sodium hydroxide solution. A paper soaked in potassium mercuric iodide solution is held at the mouth of the test tube turns brown. Which of the following could be P?

(a) Zinc sulphate
(b) Ammonium sulphate
(c) Ferrous sulphate
(d) Copper sulphate

(iv) Dry red and blue litmus paper is introduced in a solution of hydrogen chloride gas in toluene, X, Y and Z are the four observations. Which observation given below is correct?

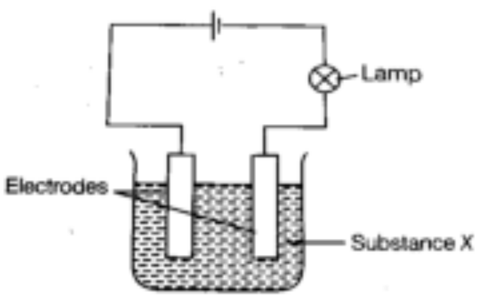
	Dry Red litmus paper	Dry Blue litmus paper
W	turns blue	turns red
X	turns pink	turns yellow
Y	turns colourless	turns pink
Z	remains red	remains blue

(a) W
(b) X
(c) Y
(d) Z

(v) Sucrose reacts with concentrated sulphuric acid to produce a black spongy mass. The reaction taking place is:

(a) Dehydration
(b) Combustion
(c) Combination
(d) Oxidation

(vi) In the circuit given below the lamp glows brightly:



What could be X?

(a) A solution of alcohol and water.
(b) A solution of sodium chloride in water.
(c) Sugar solution.
(d) Carbon tetrachloride.

(vii) The table given below provides the pH value of four solutions P, Q, R and S :

Solution	P	Q	R	S
pH	2	9	5	11

Which of the following correctly represents the solution in the increasing order of their hydrogen ion concentration?

(a) P > Q > R > S
(b) P > S > Q > R
(c) S < Q < R < P
(d) S < P < Q < R

(viii) Elements A and B forms ions A^{3+} and B^{1-}, in order to attain the stable octet electronic configuration. Elements A and B belong to the same period, their group numbers are:
(a) 1 and 3 respectively
(b) 3 and 1 respectively
(c) 17 and 13 respectively
(d) 13 and 17 respectively

(ix) A blue solution is observed when a piece of copper is added to _________ solution for half an hour.
(a) Silver nitrate
(b) Lead nitrate
(c) Zinc nitrate
(d) Magnesium nitrate

(x) The table given below shows the reaction of a few elements with acids and bases to evolve hydrogen gas. Which of the following pair of metals are amphoteric in nature?

Elements	Acid	Base
A	×	×
B	√	√
C	√	×
D	√	√

(a) A and D
(b) B and D
(c) A and C
(d) C and D

(xi) Which one of the following is NOT true with respect to conc. nitric acid?
(a) Gets a yellowish colour on standing.
(b) It is a strong oxidizing agent.
(c) Produces hydrogen on reacting with metals like Manganese and Magnesium.
(d) Volatile and corrosive

(xii) The functional group present in CH_3COCH_3 is:
(a) -O-
(b) -COOH
(c) - OH
(d) >C=O

(xiii) Formation of ionic bond is an example of:
(a) Redox Reaction
(b) Oxidation
(c) Reduction

(d) Double displacement

(xiv) The catalyst used in Contact process is:

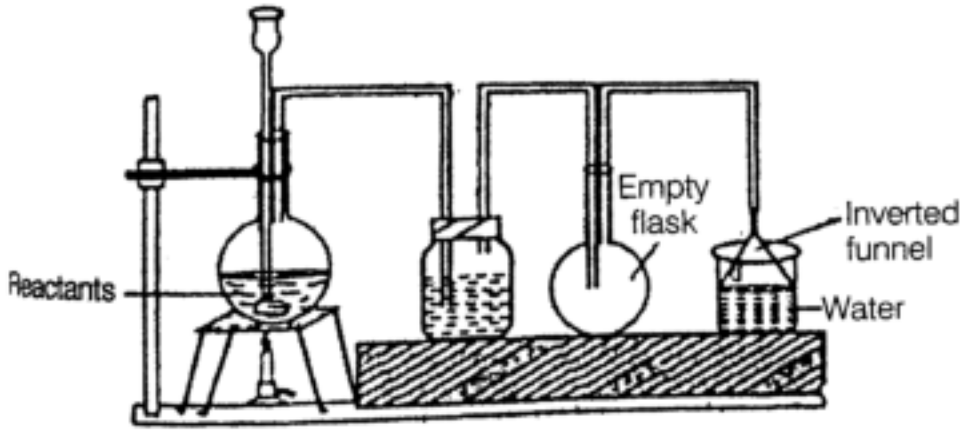
(a) Finely divided iron
(b) Vanadium pentoxide
(c) Potassium oxide
(d) Molybdenum

(xv) Which of the following is correct about an element Z with atomic number 18?

(a) It is a metalloid
(b) Highly reactive
(c) Does not form oxide
(d) It is a metal

Question 2

(i) The diagram shows the experimental setup of the laboratory preparation of an acid. Study the diagram and answer the following questions: [5]



(a) Name the acid produced.
(b) What is the role of the inverted funnel arrangement.
(c) Write balanced chemical equation for the reaction taking place in the round bottom flask.
(d) Write a chemical test to identify the acid produced above and give its relevant chemical equation.

(ii) Match the following Column A with Column B. [5]

Column A	Column B
(a) Sulphide ores	1. Non-polar covalent compound
(b) Nitric acid	2. Calcination
(c) Ammonia	3. Froth floatation
(d) Calamine	4. Ostwald's process
(e) Methane	5. Polar covalent compound

(iii) Complete the following statements by choosing the correct answer from the bracket. [5]

(a) If an element has seven electrons in the valence shell, then it is likely to have the _________ [smallest/ largest] atomic size amongst all the elements in the same period.

(b) As we descend the periodic table, the ________ [oxidizing/reducing] power of the element increases.

(c) ___________ [sulphuric acid/ hydrochloric acid] forms a normal salt and an acid salt with sodium hydroxide.

(d) A wine red colouration is observed when neutral ________ [ferrous chloride/ ferric chloride] is added to acetic acid.

(e) An __________ [acidic/ alkaline] solution will turn phenolphthalein solution pink.

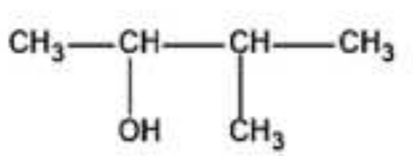
(iv) **Identify the following:** **[5]**

(a) Ions present in electrolyte which does not take part in electrode reactions.
(b) A salt formed by an incomplete neutralization of a base by an acid.
(c) Group of organic compounds having similar chemical properties in which the successive members differ by $-CH_2$.
(d) A metalloid with three electrons in its valence shell.
(e) Rocky impurities associated with ores.

(v) (a) **Draw the structural formula for the following compounds:** **[5]**

1. Ethylene
2. Pentan-2-ol
3. 2- butanone

(b) **Give the IUPAC name of the following compounds:**



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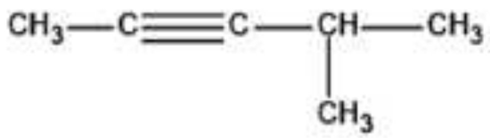

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SECTION B

(Attempt any four questions)

Question 3

(i) Identify the reactant M and write a balanced equation for the following: **[2]**
Hydrochloric acid reacts with compound M to give a salt NaCl, water and a colourless gas with a suffocating and choking odour which turns acidified potassium dichromate solution from orange to clear green.

(ii) What property of Sulphuric acid is exhibited in each of the following cases: **[2]**
(a) In the preparation of nitric acid when it reacts with potassium nitrate.
(b) When conc. Sulphuric acid is reacted with carbon to produce sulphur dioxide.

(iii) The ionization potential of an element X is greater than that of Y. **[3]**
(a) How is the reducing power of X likely to compare with that of Y?
(b) How is the electron affinity of X likely to compare with that of Y?
(c) State whether Y is likely to be placed to the left or to the right of X in the periodic table?

(iv) (a) State whether the following statements are TRUE or FALSE. Justify your answer. **[3]**
(1) When potassium chloride solution is reacted with silver nitrate solution, it produces potassium nitrate, silver and chlorine gas.
(2) Copper on reacting with nitric acid to form copper nitrate and nitrogen dioxide,

(b) Calculate the number of moles in 22 grams of nitrous oxide.
[N = 14, O = 16]

Question 4

(i) The following is an extract from 'Metals in the Services of Man Alexander and Street/Pelican 1976': [2]

'Alumina has a very melting point over 2000ºC so that it cannot readily be liquified. However, conversion of alumina to aluminium and oxygen, by electrolysis, can occur when it is dissolved in some other substance.'

(a) Name the substance which acts as a solvent for pure alumina.
(b) Give balanced equation for the reaction which takes place at anode.

(ii) Krystal heated some potassium permanganate crystals in a test tube. After collecting one litre of oxygen gas at room temperature, it was noticed that the test tube has undergone a loss in mass of 1.32 g. If 1 litre of hydrogen gas under the same condition of temperature and pressure of has a mass 0.0825 g. Calculate the relative molecular mass of oxygen. [2]

(iii) Write balanced chemical equations for each of the following: [3]
(a) Action of warm water on magnesium nitride
(b) Oxidation of Sulphur with concentrated nitric acid.
(c) Dehydrohalogenations of dibromoethane.

(iv) With respect to Haber's process answer the following: [3]
(a) Write equation for the catalyzed reaction..
(b) Calculate the volume of ammonia produced when 6 liter of hydrogen reacts with nitrogen, temperature and pressure remaining constant.
(c) State the law used for above calculation.

Question5

(i) Give reasons for the following: [2]
(a) Manish wants to prove that ammonia is a reducing agent. To demonstrate this, he reacted excess of ammonia with chlorine. What will he observe?
(b) Write a balanced chemical equation for the above reaction.

(ii) Write the function of the following components in the respective alloy: [2]
(a) Nickel in stainless steel.
(b) Magnesium in duralumin.

(iii) Rita takes a blue crystalline salt P in a test tube. On heating it produces reddish brown gas which turns acidified ferrous sulphate solution brown, a colourless, odourless gas is released which turns alkaline pyrogallol solution brown and leaves behind a black residue. [3]
(a) Name the compound P.
(b) Mention one observation when an iron nail is added to the solution of P.
(c) Identify the black residue formed.

(iv) Give reasons: [3]
(a) Burning of alkanes in limited supply of air is dangerous.
(b) Soda lime is preferred over sodium hydroxide for decarboxylation reaction.
(c) Hydrogen chloride gas fumes in moist air.

Question 6

(i) Name the following: [2]
(a) The ore of lead containing its sulphide.
(b) The substance used for concentrating bauxite.

(ii) State one observation in the following cases: [2]

(a) The gas obtained by adding warm water to calcium nitride is bubbled through a solution of potassium mercuric iodide.

(b) The gas obtained by heating rock salt with concentrated sulphuric acid is passed through lead nitrate solution and heated.

(iii) Molten lead bromide is electrolyzed using graphite anode in a silica crucible: [3]

(a) What type of reaction is taking place at the anode.

(b) Write equation for the reaction taking place at anode.

(c) Why is the process carried out in a silica crucible?

(iv) X [2,7] and Y [2,8,8,2] are two elements. Using this information complete the following: [3]

(a) ____________ is the non-metallic element.

(b) Non-metallic atoms tend to have a maximum of ___________ electrons in its valence shell.

(c) ___________ is the reducing agent.

Question 7

(i) The empirical formula of a mineral acid is H_2PO_3. Its molecular weight is 162. Calculate the mass of phosphorus in one mole of the compound.
[P = 31, O = 16, H = 1] [3]

(ii) (a) Rajesh prepared a solution Z that has a pH 7. How will the pH of solution Z change on addition of the following:

1. Acetic acid
2. Ammonium hydroxide

(b) State the position of an element having 3 shells and 8 electrons in its valence shell. [3]

(iii) Solid ammonium dichromate on heating decomposes as under: [4]

$$(NH_4)_2Cr_2O_7 \rightarrow Cr_2O_3 + 4H_2O + N_2$$

If 63 g of ammonium dichromate decomposes, calculate:

(a) The number of moles of ammonium dichromate.

(b) The mass of chromic oxide formed.

(c) The volume of nitrogen gas formed at STP.

(d) The number of moles of nitrogen formed.
[Cr = 52, N = 14, O = 16]

Question 8

(i) State giving reasons if: [2]

(a) Zinc nitrate solution and lead nitrate solution can be distinguished by adding sodium hydroxide solution.

(b) Sulphuric acid can be obtained by adding dissolving sulphur trioxide in water.

(ii) Draw the electron dot diagram of ammonium ion and label the lone pair. [2]

(iii) Give balanced chemical equations for the following: [3]

(a) Preparation of acetylene from bromoethane.

(b) Preparation of marsh gas from iodomethane.

(c) Conversion of oleum to sulphuric acid.

(iv) Identify the following substances: [3]

(a) An acidic gas formed when an alkaline gas is reacted with excess of a greenish yellow gas.

(b) The anion present in the salt which produces a dense white precipitate on addition of barium chloride, the precipitate formed is insoluble in nitric acid.

(c) The particles present in weak electrolyte.